

- 5 (a) These are the first 5 terms of a sequence.

1 8 27 64 125

Find the 10th term of this sequence.

..... 1000 [1]

- (b) These are the first 5 terms of a different sequence.

5 8 13 20 29

Find the n th term of this sequence.

$$\begin{array}{r} 5 \quad 8 \quad 13 \quad 20 \quad 29 \\ - \quad 3 \quad 5 \quad 7 \quad 9 \quad 11 \\ \hline 2 \quad 3 \quad 6 \quad 13 \quad 20 \end{array}$$

$$\begin{array}{cccccc} 5 & 8 & 13 & 20 & 29 & \underline{n^2} \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ +3 & +5 & +7 & +9 & +11 & \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ +2 & +3 & +4 & +5 & +6 & \end{array}$$

..... $n^2 + 4$ [2]

- (c) The sum of the first n terms of another sequence is $\frac{n}{2}(5n-1)$.

- (i) Use $n = 2$ to find the sum of the first two terms in this sequence.

..... 9 [1]

- (ii) Find the 3rd term of this sequence.

$$\frac{n}{2} (5n - 1) = 3$$

$$5n^2 - n = 6$$

$$5n^2 - n - 6 = 0$$

$$\frac{5n^2 - n}{2} = 3$$

$$\frac{6}{5} = 3\text{rd term}$$

..... $\frac{6}{5}$ [2]

- 6 Expand.

$$5x^2(3x-2)$$

..... $15x^3 - 10x^2$ [2]